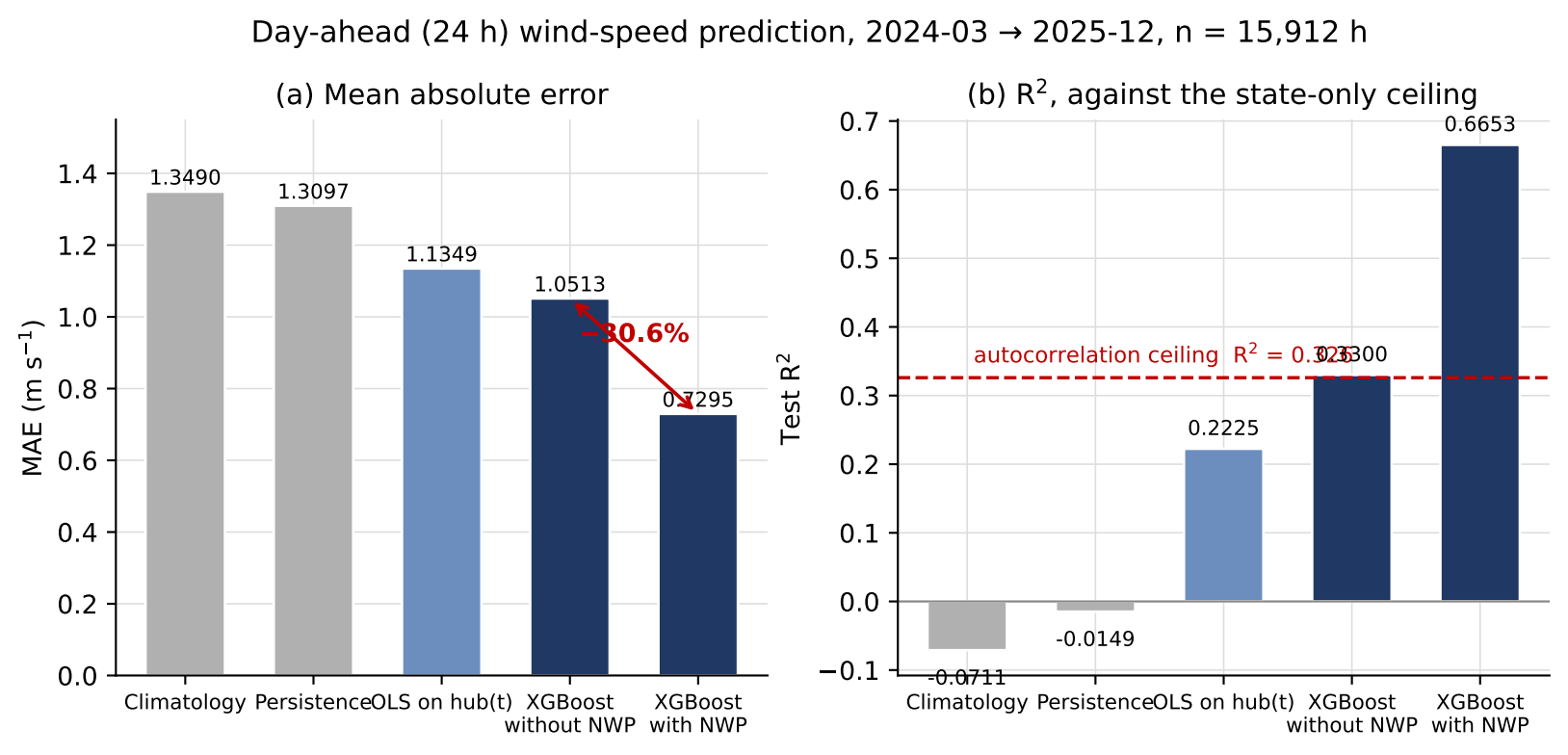
Day-ahead (24 h) wind-speed prediction, 2024-03 → 2025-12, n = 15,912 h
(a) Mean absolute error
MAE (m s^{-1})
1.3490
1.3097
1.1349
1.0513
0.7295
−30.6%
Climatology
Persistence
OLS on hub(t)
XGBoost without NWP
XGBoost with NWP
(b) R^2, against the state-only ceiling
Test R^2
autocorrelation ceiling R^2 = 0.3233
-0.0711
-0.0149
0.2225
0.3300
0.6653